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B.Tech. Degree VIII Semester Examination April 2017

CS 1801 ADVANCED ARCHITECTURE AND PARALLEL PROCESSING (2012 Scheme)

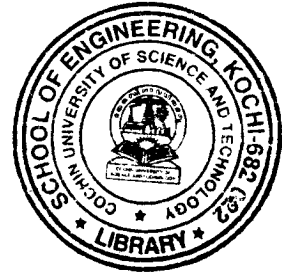
Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Write short note on PRAM.
- (b) Explain the dependences between two statements.
- (c) What are the differences between RISC and CISC architectures?
- (d) Explain the VLIW architecture. How does it differ from superscalar?
- (e) Explain about synchronous and asynchronous linear pipeline processors.
- (f) Explain the SIMD computer organization.
- (g) Write short note on PVM.
- (h) Describe thread level parallelism.



PART B

(4 × 15 = 60)

- II. (a) Explain the memory models in multiprocessor system. (9)
- (b) Explain the level of parallelism in programs. (6)
- OR**
- III. (a) Explain the different program flow mechanisms. (10)
- (b) What do you mean by Amdahl's Law? Explain briefly. (5)
- IV. (a) Explain memory hierarchy technology. (10)
- (b) Briefly explain the shared memory organization. (5)
- OR**
- V. (a) Explain back plane bus systems and operational principles of bus. (10)
- (b) Explain the sequential and weak consistency models in shared memory. (5)

(P.T.O.)

VI. Consider the following reservation table for a three stage pipeline. (15)

	1	2	3	4
S1	X			X
S2		X		
S3			X	

- What are the forbidden latencies?
- Draw the state transition diagram.
- List all the simple cycles and greedy cycles.
- Determine the optimal constant latency cycle and MAL.
- Determine the throughput of this pipeline.

OR

- VII. (a) Explain Cache coherence and synchronization mechanism. (10)
- (b) Explain Linear and Non-linear pipelines. (5)

VIII. Explain the parallel programming models. (15)

OR

- IX. (a) Describe the Dependence analysis of data arrays. (7)
- (b) Explain Tomusulo's algorithm. (8)